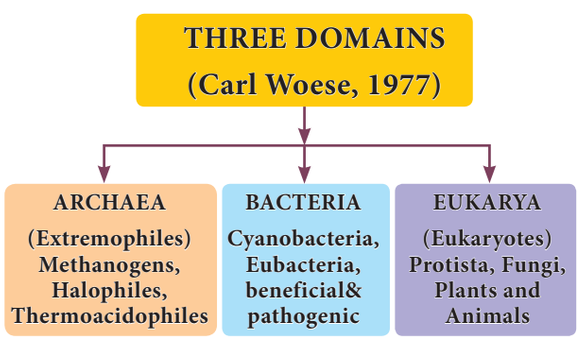
characters produced a **phylogenetic tree or cladogram** (Figure 1.1).

Depending on the system of classification, organisms were classified into two or three kingdoms. Later into four, five, six and now into seven kingdoms. **R.H.Whittaker** (1969) proposed the **Five kingdom Classification**, the Kingdoms defined by him were **Monera, Protista, Fungi, Plantae, and Animalia** based on the cell structure, mode of nutrition, mode of reproduction and phylogenetic relationships. Table 1. gives a comparative account of different characteristics of the five kingdoms.

Classification has come a long way and now takes into an account even molecular level DNA and RNA identification. The advancement in molecular techniques and biochemical assays has led to a new classification - The "**Three Domain" classification**.

1.4 Three Domains of life

Three domain classification was proposed by **Carl Woese** (1977) and his co-workers. They classified organisms based on the difference in 16S rRNA genes. The three domain system adds the taxon 'domain' higher than the kingdom. This system emphasizes the separation of Prokaryotes into two domains, Bacteria and Archaea, and all the eukaryotes are placed into the domain Eukarya. Archaea appears to have more in common with the Eukarya than the Bacteria. Archaea differ from bacteria in cell wall composition and differs from bacteria and eukaryotes in membrane composition and rRNA types.



1. Domain Archaea

This domain includes single celled organisms, the prokaryotes which have the ability to grow in extreme conditions like volcano vents, hot springs and polar ice caps, hence are also called **extremophiles.** They are capable of synthesizing their food without sunlight and oxygen by utilizing hydrogen sulphide and other chemicals from the volcanic vents. Some of the them produced methane (methanogens), few live in salty environments (Halophiles) and are thermoacidophiles which thrive in acidic environments and at high temperatures.



Thermus aquatics is a bacterium which can tolerate high temperatures. The first DNA polymerase enzyme was isolated from *T. aquaticus* it is used in **PCR** (Polymerase Chain Reaction) for DNA amplification.

2. Domain Bacteria

Bacteria are prokaryotic, their cells have no definite nucleus and DNA exists as a circular chromosomes and do not have histones associated with it. They do not possess membrane bound organelles except for ribosome (70S type). Their cell wall contains peptidoglycans. Many are decomposers, some are photo-synthesizers and few cause diseases. There are beneficial **probiotic** bacteria

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